SEMINAR III · WEEK 7

Letters That Stand for Numbers

A variable isn't a mystery to be solved every time you see one — sometimes it's just a value that changes, like hours worked or number of items bought. This week is about reading, building, and simplifying expressions that use them.

What Is a Variable?

A **variable** is a letter standing in for a number that's unknown or that changes — hours worked, number of tickets, temperature. A **constant** is a fixed value that doesn't change. In $3x + 7$, the 3 is a **coefficient** (it multiplies the variable), x is the variable, and 7 is the constant.

> A variable doesn't always mean "find this missing number." Plenty of expressions describe a changing quantity and are never meant to be solved for x.

Reading and Writing Expressions

Translate carefully between words and symbols in both directions.

> **Watch the order:** "five less than a number" means $x-5$, **not** $5-x$. "Less than" reverses the order you might expect.

Substituting Into an Expression

Replace *every* occurrence of the variable with its given value, then follow the order of operations exactly as usual — substitution doesn't change that rule.

Evaluate $5x - 3$ when $x = 4$: $5(4) - 3 = 20 - 3 =$ **17**.

In $3x + 2x$, both x's get replaced when you substitute — not just one.

Like Terms

Terms combine only when they share the same variable raised to the same power. $2x$ and $3x$ combine to $5x$. $4x$ and $3x^2$ do **not** combine — different powers, not like terms.

The Distributive Property

The outside factor multiplies *every* term inside the parentheses: $3(x+2) = 3x+6$, not $3x+2$.

A negative factor distributes too: $-2(x+6) = -2x-12$, not $-2x+6$.

Putting It Together

Simplify $3(x + 2) + 4x$: distribute first: $3x+6+4x$. Then combine like terms: $\mathbf{7x + 6}$.

Expressions in Real Life

A weekly wage with a bonus, a car rental with a per-day fee, a subscription with a base price plus usage — all of these translate into expressions built from a variable, a coefficient, and a constant.

"Twelve more than twice a number": $2x + 12$.

Does My Answer Make Sense?

Check	Ask
Size	Is the result approximately what I expected?

Sign	Should it be positive or negative?
Substitution	Did I replace every occurrence of the variable?
Order of Operations	Did I still follow it after substituting?

Error Detective

Knowledge — "I don't remember what a coefficient is."

Fix: review the vocabulary — coefficient, variable, constant, term.

Process — distributing to only one term

3(x+4) becomes 3x+4 instead of 3x+12. Fix: multiply the outside factor by *every* term inside.

Execution — correct setup, arithmetic slip

The right terms were combined, but a sign or digit was copied wrong along the way. Fix: check the work line by line.

Comprehension — "five less than a number" written as 5−x

The phrase was translated in the wrong order. Fix: reread the phrase and identify which quantity comes first.

Strategy — substituting before simplifying

Substituting into a messy, uncombined expression makes arithmetic harder than it needs to be. Fix: simplify (combine like terms, distribute) before substituting when possible.

Quick Reference

Variable: a symbol representing an unknown or changing value.

Constant: a fixed value.

Coefficient: a number multiplied by a variable.

Like terms: same variable, same power — only then can they combine.

Distribute: multiply the outside factor by every term inside.

Check: Size → Sign → Substitution → Order of Operations.

Keep this page. You'll use it all year.